UNIVERSITY OF COMPUTER SCIENCE & ENGINEERING
End Semester Examination - November 2025
Subject: Database Management Systems (CS-302)

Time Allowed: 3 Hours Maximum Marks: 100

Instructions:
1. Section A is compulsory (Answer all questions).
2. Section B: Answer any FOUR questions out of six.
3. Figures to the right indicate full marks.

SECTION A (20 Marks)
Answer all questions. Each carries 5 marks.

Q1. Explain the three-level ANSI/SPARC database architecture with a neat diagram. [5 marks]

Q2. Define Super Key, Candidate Key, and Primary Key with suitable examples. [5 marks]

Q3. What are ACID properties? Explain the significance of Durability in DBMS. [5 marks]

Q4. Differentiate between file-based systems and DBMS. [5 marks]

SECTION B (80 Marks)
Answer any FOUR questions. Each question carries 20 marks.

Q5. (a) What is Normalization? Explain 1NF, 2NF, and 3NF with examples. [10 marks]
(b) When is a relation considered in BCNF? Differentiate 3NF from BCNF. [10 marks]

SECTION B (Continued)

Q6. Construct an Entity-Relationship (ER) diagram for an Online University Library Management System. Specify entity sets, r

Q7. (a) What is Relational Algebra? Explain SELECT, PROJECT, and CARTESIAN PRODUCT operations. [10 marks]
(b) Write SQL queries for a relation Student(RollNo, Name, Department, Marks):
(i) List all students with marks greater than 75 in CSE. [5 marks]
(ii) Find the average marks per department. [5 marks]

Q8. Explain the Two-Phase Locking (2PL) protocol. How does it prevent concurrency anomalies? Differentiate between Strict